

L10: Submodularity of Objective Function

$A \subseteq B$



$$f(A \cup \{s\}) - f(A) \geq f(B \cup \{s\}) - f(B)$$

The function we try to maximize, $f(S)$, is nonlinear and it obviously depends on the topology of the network as well as on the specific diffusion model.

As we will see at the next page, however, the function $f(S)$ has an interesting property referred to as “**submodularity**”.

In this page, we first define submodularity – and then state an important result on how to optimize such functions.

Suppose that a function $f(X)$ maps a finite subset X of elements from a ground set U to non-negative real numbers.

The function $f(X)$ is called **monotone** if $f(v \cup X) \geq f(X)$ for any $v \in U$ and any subset X .

The function $f(X)$ is called **submodular** if it satisfies the following “**diminishing returns**” property:

$f(v \cup X) - f(X) \geq f(v \cup T) - f(T)$ for any subsets X and T of U such that $X \subset T$.

In other words, the marginal gain from adding an element v to a set X is at least as high as the marginal gain from adding the same element to a superset T of X .

Dr. Nemhauser (a professor at Georgia Tech) and his colleagues proved the following facts about monotone and submodular functions:

- The bad news: optimizing such functions, subject to the constraint that X includes k elements, is an NP-Hard problem, and so it cannot be solved efficiently by any algorithm (unless if $P=NP$).
- The good news: Consider an iterative greedy heuristic that adds one element in X in each iteration, selecting the element that gives the maximum increase in $f(X)$. This simple algorithm has an approximation ratio $(1 - \frac{1}{e})$. This means that the greedy solution cannot be worse than about 63% of the optimum value of $f(X)$.

The previous result is very important in practice because it means that, even though the greedy algorithm is suboptimal, we have a reasonable bound on its distance from the optimal solution.

Food For Thought

Prove that a non-negative linear combination of a set of submodular functions is also a submodular function.